

ure through workload redistribution across a clustered solution. In addition, it could recover data almost instantaneously through data snapshots. With these safeguards in place, companies can enjoy the assurance of 99.999 per cent availability.

High scalability. Enterprises also need to consider future business and data growth in their selection of storage solutions. For example, Shanghai Ocean University was looking to manage 100 terabytes of e-books and videos within its digital library on campus while also planning ahead for its role as a super-computing centre. With scalability and flexibility in mind, it chose a unified storage platform that was not only a high-performance file serving solution but also scalable up to 15 petabytes in the future. Dynamic storage tiering (DST) within this platform ensured the most efficient use of storage locations for the university's data.

DST enables data to dynamically move to different storage tiers depending on usage needs. For example, frequently accessed data would be stored on tier-one servers that consist of faster disks, while less volatile data resides on tier-two or tier-three servers. Because most organisations access only 10 per cent of their data frequently, 90 per cent of the university's library data would be redirected to the less expensive storage using DST technology, which is expected to bring more than \$700,000 in savings.

High performance. The constant demand for data is putting a strain on

Drowning in oceans of data

The increase in data creation is taking place at personal and enterprise levels. At the personal level, the rise of personal digital devices speaks volumes about the way we create and share personal data. Photos have moved from physical photo albums to things like smartphones and photo-sharing websites. On Facebook alone, more than 25 billion pieces of content are shared each month, requiring over 60,000 servers to handle the storage and transactions.

At the enterprise level, there is unprecedented growth in online transactions and services that need to be stored, secured and managed. For companies like Taobao, an e-commerce site in China comparable to eBay, reliable data storage is critical, considering that in one day in September 2009 the company sold 4.561 million items and processed 64,808 transactions. With each item holding an accompanying description, photos and customer data, there is a burgeoning need for robust and scalable storage solutions.

storage system performance. To meet this demand, there is a need for even utilisation across multiple nodes within a single storage system. A unified storage system addresses this by enabling every NAS engine node to share the same storage pool simultaneously. In addition, it could make it possible for multiple NAS engines to read and write the same document simultaneously, thus improving response time and performance when many users are trying to access a single document.

Management simplicity. While technologies are becoming more sophisticated, their management has to be simple and intuitive. Unified storage systems offer centralised management tools that are easy to use and access, whether through a graphical user interface application or a Web interface. System administrators can use these management tools to set notification alerts via multiple communication mechanisms in the case of

a system failure. Data backups can be configured and monitored for success and failures.

In short

The digital revolution continues unabated, and there will be more changes in data creation, security, storage and management as we move into a digital universe of unfathomable amounts of content. For enterprises, the demand for innovative storage solutions will only increase with time and it will become increasingly important for these solutions to be highly reliable, scalable and simple to manage.

Unified storage platforms can adapt to the changing business needs of enterprises, provide them with the peace of mind to focus on improving the business, just as Taobao and Shanghai Ocean University have. ●

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